

Shubhayu Majumdar

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SKILLS SUMMARY

- **Languages:** Python, Javascript, C/C++, Dart
- **Frameworks:** FastAPI, Flask, Flutter, PyTorch, TensorFlow
- **AI / ML:** Gemini, Vertex AI, RAG, LangChain, Multimodal GenAI, Embeddings
- **Databases:** MySQL, MongoDB, PostgreSQL, SQLite, Firebase
- **Platforms:** Google Cloud Platform, Linux, GitHub, Raspberry Pi

EXPERIENCE

- **BlueVector AI** Kolkata, India
Cloud Developer July 2024 - Present
 - Architected and deployed a **production RAG pipeline** for the Iowa Department of Revenue, built a retrieval corpus of **59,346 tax code and 91,338 administrative code extracts** embedded into **Vertex AI Datastore**, with a **self-validating reranking pipeline** generating legally-cited determinations with near-perfect accuracy against DOR test cases. Successfully delivered to **UAT and production**.
 - Architected a **config-driven multi-database layer** supporting **MySQL, MongoDB, and PostgreSQL** implemented a **factory pattern** for cross-DB CRUD and cross-source entity references, keeping routing transparent and clean.
 - Sole owner and engineer of **AI Core Features**: built and maintained the company's entire AI layer across the **full suite of Gemini and Vertex AI capabilities**, including multimodal generation, **custom Vertex AI Datastore groundings**, semantic search, and **multiple RAG and GenAI pipelines** across client projects.
 - Built a **multimodal GenAI record generation** feature using table schema, field types, and user prompts to generate fully-populated records including **cross-referenced child entities**; heavily used for client prototyping and demos.
 - Built a **custom CLI tool** for multi-environment management automating deployment of app config, user data, and **product documentation** across environments, replacing error-prone manual processes.
 - Resolved **200+ on-call issues** leveraging **GCP logging and observability services** for debugging and RCA, drove a **30% improvement in incident response time** through systematic root cause analysis and process improvements.
 - Grew from early-stage engineer to **platform owner across core systems**, contributed to SnapApp through its **full lifecycle from inception to market launch**, owning AI, database architecture, developer tooling, and reliability.
- **BlueVector AI** Kolkata, India
Associate Cloud Developer May 2023 - June 2024
 - Integrated **Document AI** for bulk ingestion into SnapApp with custom and pre-built parsers, reduced **import time by 90%** for new parsers.
 - Conducted **full VAPT** across authentication, API, and input validation layers, patched **8 critical SQLi/HTMLi vulnerabilities** and resolved **200+ high/critical SonarQube findings**, eliminating attack surface across services.
- **Indian Institute of Technology Kharagpur** Kharagpur, India
Machine Learning Research Intern May 2022 - Aug 2022
 - **ML Research:** Built a custom scraper to collect **35,600 user records and 73,000 images**, training a **two-branch CNN+MLP** model on the dataset to predict Instagram post popularity with **82% accuracy**.

EDUCATION

- **Techno Main Saltlake** Kolkata, India
B.Tech in Computer Science (Artificial Intelligence and Machine Learning) Nov 2020 - Jun 2024
GPA: **8.86**

PROJECTS

- **ConnecTen (Flutter, Dart, Firebase, Nearby Connections API)**: Mobile application to share social links and build connections seamlessly with nearby people **without using internet** targeted for meetups and gatherings. Won **Innervive 7.0** by AIT Pune, **HackNITR 4.0** by NIT Rourkela, **Hackday** by GDGC Kolkata, CIEM Hackathon.
- **Envisage App (Python, FastAPI, Dart, Flutter, Firebase, Cashfree)**: Research-oriented, full-stack, event management app with an integrated payment gateway that impacted more than **1200 users**, bringing a **30% increase** in participation.
- **Discord Study Bot (Web Development, Discord API)**: Open source, discord bot to record study time of members and make a leaderboard to increase engagement. The bot is currently used in the discord server of **IIT Hyderabad** with more than **6000 users**. Tech: **Python, Discord API, MongoDB, Heroku**.

HONORS AND AWARDS

- Awarded **Swiggy Bug Bounty Hall of Fame**. Acknowledged by **NCIIPC, Govt of India** and **NCSC, Govt of UK**.
- **Certified Network Security Specialist** from **ICSI, UK**.
- Published information and descriptive exploits about **several vulnerabilities** and helped secure multiple companies.

COMMUNITY AND LEADERSHIP

- **TFUG Kolkata — Organiser | GDG Cloud Kolkata — Core Team and Speaker** Kolkata, India
Organising AI/ML events across a 11,500+ developer ecosystem 2022 - Present
- **Pan-India Speaker — AI/ML, GCP and Backend** India
Delivered 30+ talks and workshops across developer communities nationwide 2022 - Present